

ITIS5000V

Intro to Data Science

Module 7: Project Report

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Predicting Churn for Better Retention: Using Machine Learning to Predict Likely Customer Churn and to Develop Retention Strategies in the Retail Banking Industry.

I. ABSTRACT

Abstract – This study evaluates the degree to which machine-learning models can predict retail-banking customer churn and subsequently support targeted retention strategies. Six models were trained on a public dataset of 10,000 European banking customers using an 80/20 train-test split, stratified five-fold validation cross-validation and SMOTE applied to balance the training data. CatBoost achieved the strongest ROC-AUC while still maintaining competitive recall and F1 Performance. Meanwhile, LightGBM showed the greatest results in the Cross-validated F1 score, slightly ahead of CatBoost and XGBoost. SHAP analysis identified age, product number and active member status as the most influential predictors of churn with account balance and tenure also showing some effect. The results reveal that examining accuracy alone can obscure weak identification of churners and model selection should instead reflect the cost of missing churners versus unnecessary churn intervention actions. A churn-risk and estimated-value matrix is ultimately proposed to translate the model outputs into differentiated retention action.

I. INTRODUCTION

Customer churn is a constant issue in the banking industry that can have a significant impact on the profitability of a business. Churn occurs when customers reduce or end their relationship with the business whether by closing accounts, reducing their account balances or failing to renew certain products. The ultimate effect of churn is the loss of future revenue from those customers leaving the business but also the costs of acquiring new customers to replace them whether through marketing expenses, onboarding costs or other administrative charges. As a consequence, reducing customer churn is an important aspect of maintaining profitability as an organization. Increasing customer retention by 5% has the potential to increase profits by more than 25% (Reichheld, 2001).

However, not all customers will have equal value to the business and as such blanket retention campaigns can come with unnecessary wastage of resources.

To address this, we can examine the substantial amounts of data that banks hold on their customer base, including balances, product usage, transaction activity, engagement and demographic information. Building a machine learning model that can use these variables to identify patterns that are associated with departing customers will allow the business to begin to target their retention efforts towards customers that are exhibiting signs of potential churn and determining the level of resources that should be directed towards maintaining their business.

This project, therefore, will attempt to determine to what extent machine-learning models can use banking customer data to predict customer churn and how can these predictions be used to support targeted retention strategies in retail banking. The most useful model here will not necessarily be the one that has the greatest accuracy but the one that can provide manageable grouping of customers at risk of churn and understandable basis for action that managers can use to apply appropriate retention action.

II. PREVIOUS WORK

Given the significance of minimising customer churn to profitability, there is a significant basis of established work upon which to build. Farquhar (2004) found that while financial institutions recognised the criticality of customer retention, they sometimes relied on systems that were focused on individual transactions rather than the long-term relationship.

Farquhar and Panther (2008) later found that often, banks struggled to balance the acquiring new customers with the retention of profitable existing relationships. Their results suggest that it is more

valuable to target retention efforts towards those customers which provide high value rather than attempting a blanket strategy of retaining all customers.

The reasons for why customers leave banks have also been examined widely, with Colgate and Hedge (2001) researched this in the Australia and New Zealand banking sector, classifying three main problem areas of 'service failures, pricing problems and denied services.' They found that problems with pricing was the factor with the most important impact on customers switching to other banks. However, they also found that customers tended to complain more about service failures they had experienced prior to leaving the bank, which suggested that customers may be remaining silent on the problems that were impactful for their decision to leave the bank. Therefore, examining behavioural changes may provide a better early warning system than looking at complaints alone.

With regard to studies that have previously applied models to the examination of customer churn data, Van den Poel and Larivière (2004) combined several churn predictors into a single retention model related to customer behaviour and found that demographic characteristics and environmental changes were a large concern when examining churn but that 'stimulating interactive and continuous relationships' was another important factor. Conversely, their work suggested that behavioural predictors had only a limited impact on attrition.

Later, Larivière and Van de Poel (2005) used data from a real-life sample of 100,000 banking customers to compare random forests with traditional regression techniques to predict customer retention and profitability. They found that random forests provided a better fit for predicting purchasing, churn and profitability than logistic and linear regression. Their results also found that the predictors of churn were not one-to-one with the predictors of profitability. This would support the conclusion that churn risk should be evaluated separately from customer value as while one customer with high churn risk may present limited future value, another with similar churn risk may be a highly profitable customer.

Work in this area such as Ekinci, Uray and Ülengin (2014), have developed models for determining customer lifetime value (CLV). Their work predicted the potential value of current customers through examining monetary activity, product ownership and risk of certain bank services,

rather than measuring the current value. These studies form a basis for combining churn risk with measures of future profitability when determining which customers should be targeted for retention measures.

In more recent years, there have been studies that have compared various machine-learning algorithms for predicting bank churn. In Tékouabou et. (2022), which highlighted the importance of addressing class imbalances and explanations of complex models found that Random Forest was among the most effective algorithms in terms of both accuracy and F1 score. Singh et al. (2024) compared multiple machine learning algorithms while developing a Data Visualisation RShiny app for management analysis that found that Random Forest and XGBoost delivered strong predictive performance. They also found that creating usable visualisations of the results were vital to supporting management decisions based on the data.

Taken all together, the research suggests that successful management of churn requires appropriate handling of imbalanced data, easily explainable predictions, an evaluation of customer-value and providing a clear link between the specific churn drivers and possible retention actions.

III. METHODOLOGY

This project used the CRISP-DM framework and a publicly available retail-banking churn dataset from Kaggle that contained 10,000 European banking customers and 12 individual variables. The target variable was of course, whether a customer had churned while the surrounding predictor information comprised credit score, gender, tenure, account balance, total banking products, age, credit-card ownership, salary and active-member status. There were no missing values, but Customer ID was excluded as it was an identifier with meaningful prediction value.

First, the exploratory data analysis was performed using Python, Pandas, Seaborn and Matplotlib. From this, churn distributions, categorical churn rates, boxplots, descriptive statistics and correlations were reviewed to identify relationships and any data-quality issues. In total, the dataset included 7,963 customers who retained a relationship with the bank and 2,037 customer who churned, representing approximately a 20% churn rate. This imbalance could, of course present problems with using model accuracy rates alone, as this could

provide a misleading assessment of the model quality.

For predictors, gender was encoded as a binary variable, while country used one-hot encoding. Six predictors (credit score, age, tenure, balance, number of products and estimated salary) were continuous and so, standardised using 'StandardScaler'. The data was separated into an 80/20 split of training to testing data, representing 8,000 training observations and 2,000 testing observations. As the churn class was underrepresented to a significant degree, Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data. This increased the training observations from 6,356 non-churn customer against 1,644 churners to 6,356 observations in each class.

Six classification algorithms were then evaluated: Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM and CatBoost. Performance of these models was evaluated using accuracy, precision, recall, F1-score and ROC-AUC.

In order to reduce the dependence on one train-test split, a stratified five-fold cross-validation procedure was conducted on the training observations. Pre-processing and SMOTE were incorporated into the cross-validation pipeline so that both scaling and synthetic observations were generated independently and thus reduce the risk of data leakage (Figure 1).

The CatBoost model was selected for further interpretation due to it achieving the highest mean cross-validated ROC-AUC while also maintaining strong recall and F1 performance. To evaluate the overall importance and direction of customer characteristics that affect churn predictions, SHapley Additive exPlanations (SHAP) analysis was used.

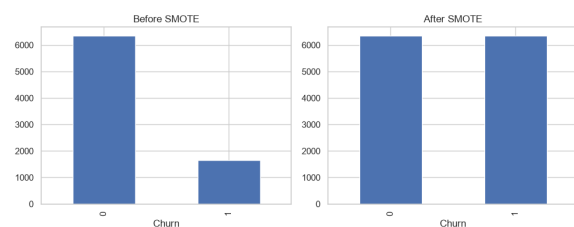


Figure 1 – Results of Applying SMOTE

Finally, the retention framework combined the predicted churn risk with the estimated customer value. Unfortunately, actual lifetime value and profitability values were unavailable, so a weighted index was used that consisted of 50% account balance, 30% number of products and 20% active-member status. Following this, customers were categorised according to their churn risk and their estimated value to illustrate differentiated retention strategies.

IV. RESULTS

The exploratory analysis identified a number of important differences between churned customer and those that remained with the bank. Customer that churned tended to be older (an average of 44.84 years versus 37.41 for those that remained) and maintained a higher average account balance (approximately \$91,100 versus \$72,700 for remaining customers). Comparatively, credit score, tenure and salary showed much smaller differences.

Another predictor showing a strong relationship with churn was customer engagement where around 26.85% of inactive customers churned versus just 14.27% of active customers suggesting that active-member status could provide a strong retention signal.

The Logistic Regression model was used as a baseline for evaluating the models tested and achieved an 81.15% accuracy and 20.10% recall score. The models confusion matrix showed that it correctly identified just 79 of 393 churners in the testing sample. Therefore, we can see that the strong accuracy concealed poor performance in the more valuable class of managerial interest. Once SMOTE was applied however, recall increased to 70.74% while accuracy fell to 71.25%. F1-score saw a significant increase while precision saw a decrease. When applying this retention, increasing identification of customers at risk may be a worthwhile trade-off for accepting more false-positive contacts.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	70.80%	38.25%	68.25%	49.00%	76.52%
Decision Tree	75.81%	44.73%	67.22%	53.42%	80.22%
Random Forest	84.91%	65.69%	55.78%	60.31%	84.89%
XGBoost	84.79%	63.72%	60.89%	62.24%	86.08%
LightGBM	85.25%	65.25%	60.71%	62.87%	86.21%
CatBoost	84.07%	60.49%	64.96%	62.62%	86.59%

Figure 2 – Model Comparison of Cross-Validation Results Across All Evaluation Criteria

The table in Figure 2 presents the results of each model across the evaluation criteria used and here we can see the trade-offs that managers must consider. LightGBM achieved both the greatest Accuracy (85.25%) and F1-score (62.87%). Random Forest produced the highest Precision (65.69%) while Logistic Regression presented the highest Recall (68.25%) with a much lower precision and F1-score. CatBoost posted the highest ROC-AUC at 86.59% and higher recall than LightGBM, demonstrating a strong and consistent ability to distinguish customers who churn from those who remain. CatBoost was then selected as the primary model for explainability and customer segmentation because it achieved the highest ROC-AUC, competitive F1-score, and stronger Recall, aligning well with the project's customer-retention objective. Taking the Cross-Validated F1-Score as an overarching rating for the models, we can see that the consistency of the XGBoost, LightGBM and CatBoost models demonstrates the advantage of using those models capable of analysis complex, non-linear relationships (Figure 3).

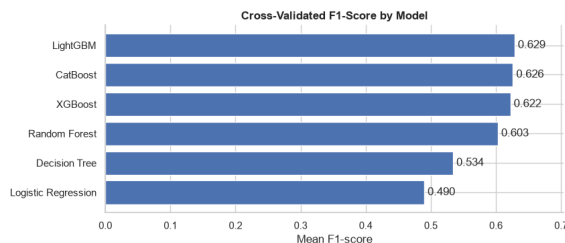


Figure 3 – Cross Validated F1-Score by Model

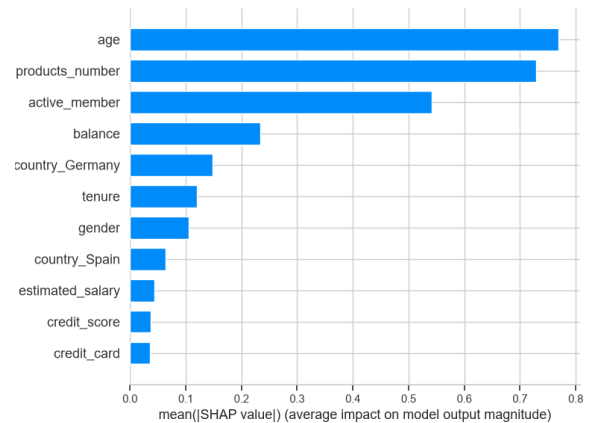


Figure 4 – SHAP Feature Importance

The subsequent SHAP (Figure 4) analysis identified, as stated earlier, that age was the most influential predictor of customer churn with older, particularly middle-aged customers presenting increased churn risk, though this effect declined amongst the oldest customers. This predictor was then followed by number of banking products, active membership, account balance and tenure. Customers holding two products generally showed a reduced churn risk than those with one product however those holding three or four products showed the greatest churn prediction which suggests that product ownership does not directly translate into greater retention (Figure 5).

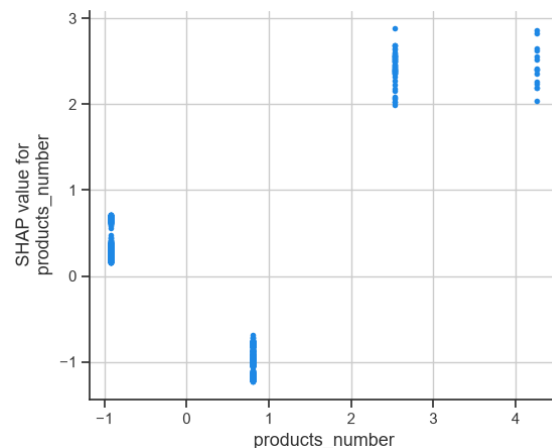


Figure 5 – Customer Churn by Number of Products

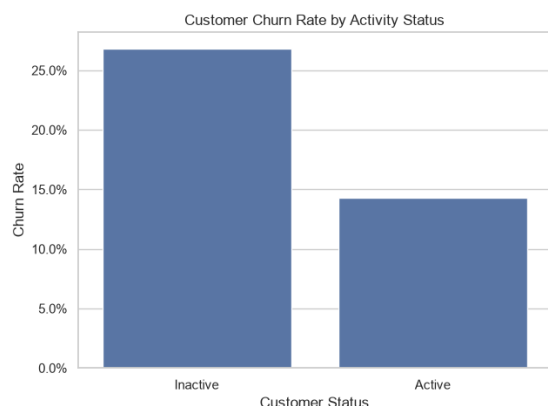


Figure 6 – Customer Churn by Activity Status

Active membership was consistent in its reduced predicted churn probability, consistent with the exploratory data analysis which indicates that customer engagement is associated with retention (Figure 6).

V. DISCUSSION

The results present strong evidence that machine-learning models can be used to identify meaningful patterns in retail-banking churn and also that model selection should be reflective of the business objective and not be predicated simply on the model with the highest accuracy.

Applying SMOTE significantly improved the models ability to identify at-risk customers however this came at the expense of increased false positives. This trade-off should be a consideration for managers in the banking sector when evaluating whether missing a high-value customer that eventually leaves is more costly than engaging customers who would have remained regardless. CatBoost was selected as the primary model for this analysis however, the preferred algorithm for managers will depend on manager prioritisation, whether that's minimising wasted interventions or identifying churners.

The SHAP analysis particularly is important for managers as it helps to identify the particular characteristics of potential churners. For example, with inactive customers churning significantly more than active customers, declining customer engagement could spur banks to increase communication or suggest personalised discussions. Interesting to note is that higher number of products did not translate to reduced risk, with customers holding three or four products showing increased churn risk suggesting cross-selling may not be

considered a positive strategy versus gaining new customers.

Regarding limitations, SMOTE analysis was useful in managing the imbalance in the data, however these synthetic observations are imperfect representations of actual customer behaviour. The 50%/40%/20% customer weighting according to balance, products and activity is an approximation, rather than a profitability model that has been validated. In future analysis, this could be replaced with a more appropriate split. In addition, the use of a dataset of European banking customers may limit its general applicability to the Canadian banking sector.

Applying these results practically will also require analysis of retention techniques to determine not just which customers are likely to churn but also, which customers it is possible to retain, what retention technique is most applicable and what is the financial return or cost of the applied retention, analysed in the Customer Retention Priority Matrix (Exhibit F).

Machine-learning can be a useful tool in providing support for customer-retention action however its greatest value will come from providing strong prediction combined with customer value and retention technique analysis to provide the data needed for appropriate managerial judgement to be applied to available retention resources.

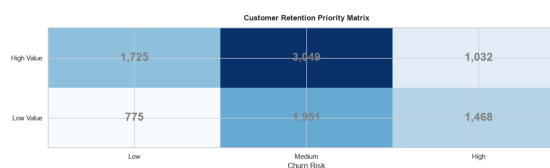


Figure 7 – Customer Retention Priority Matrix

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Appendix

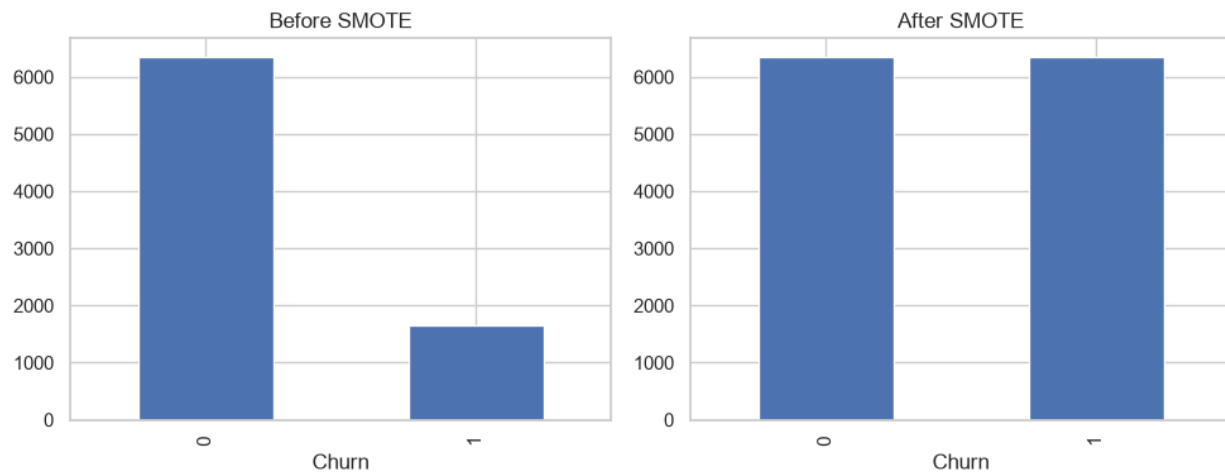


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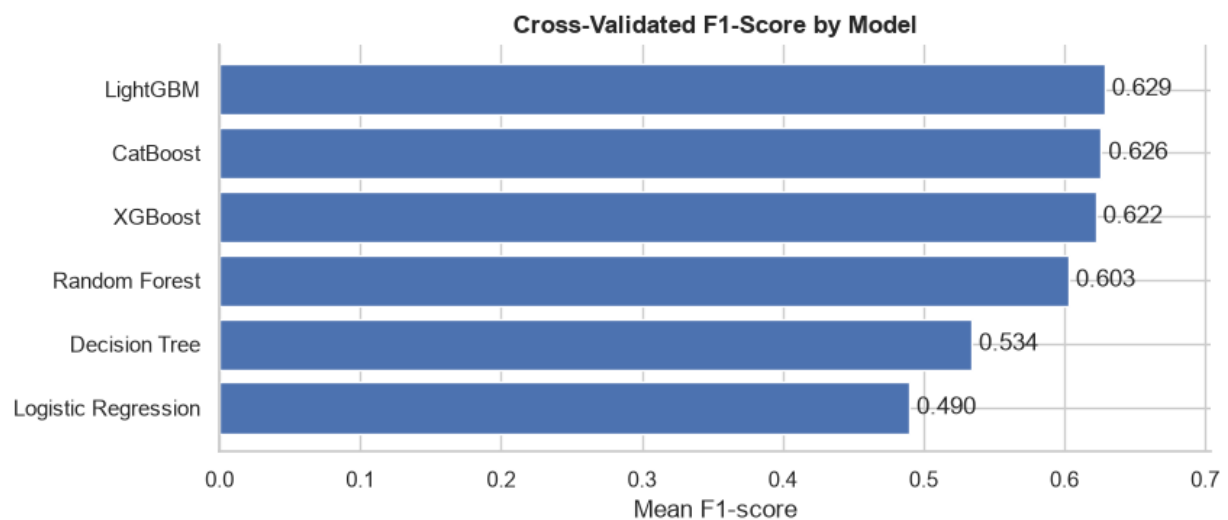


Figure 3 – Cross Validated F1-Score by Model

Appendix

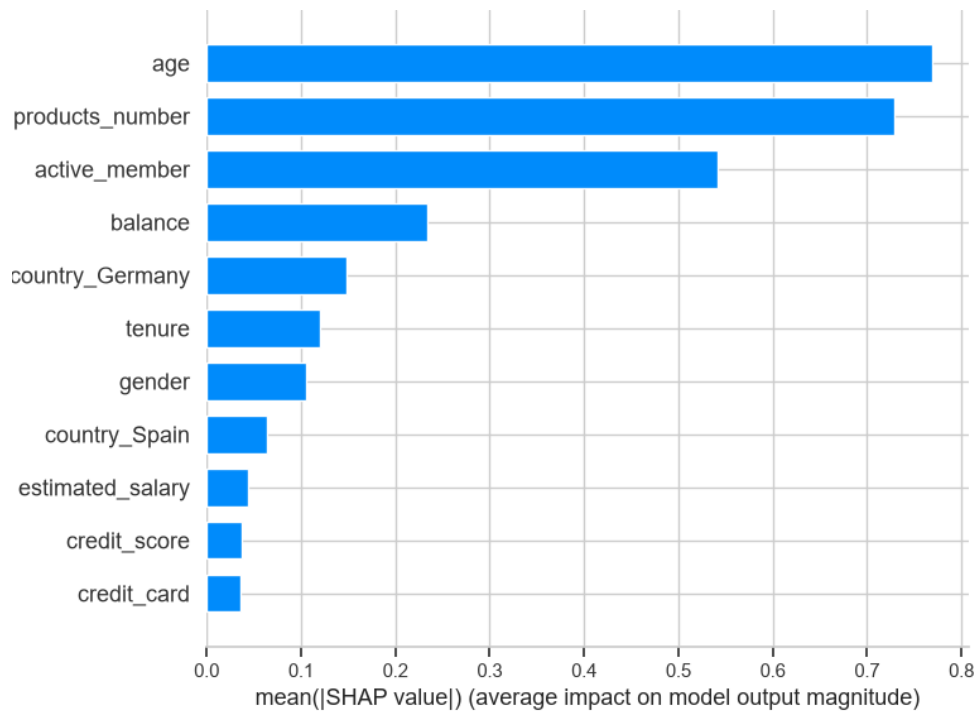


Figure 4 – SHAP feature importance

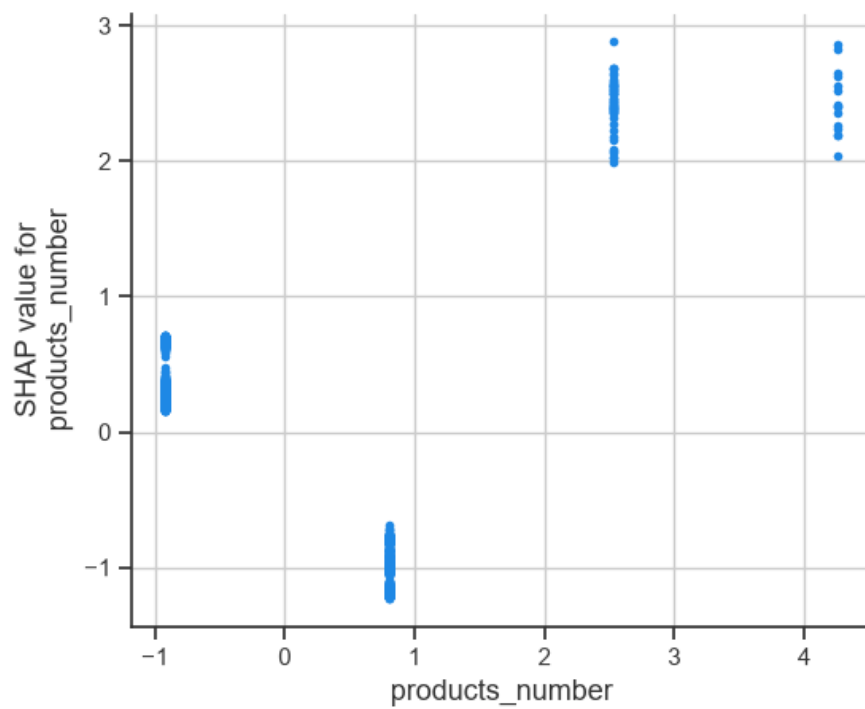


Figure 5 – Customer Churn Rate by Number of Products Held

Appendix

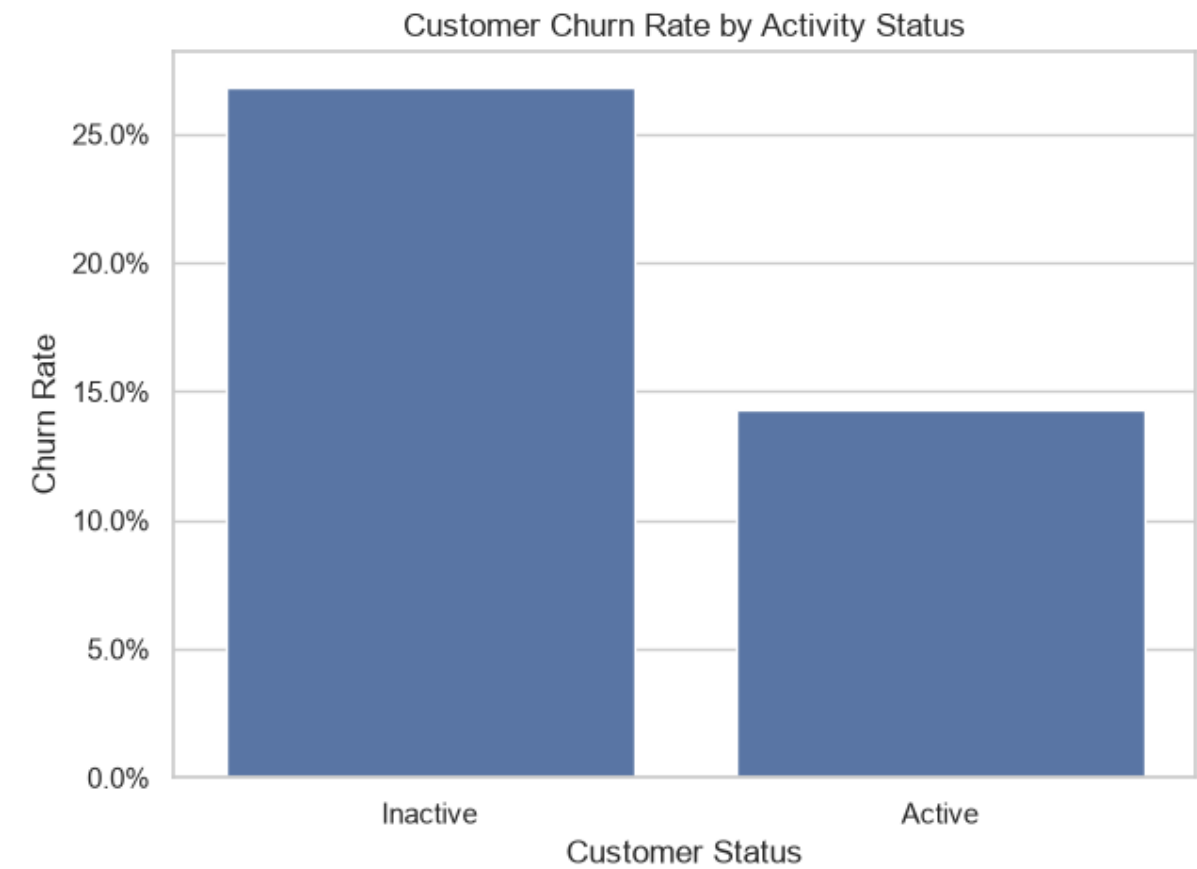


Figure 6 – Customer Churn by Activity Status

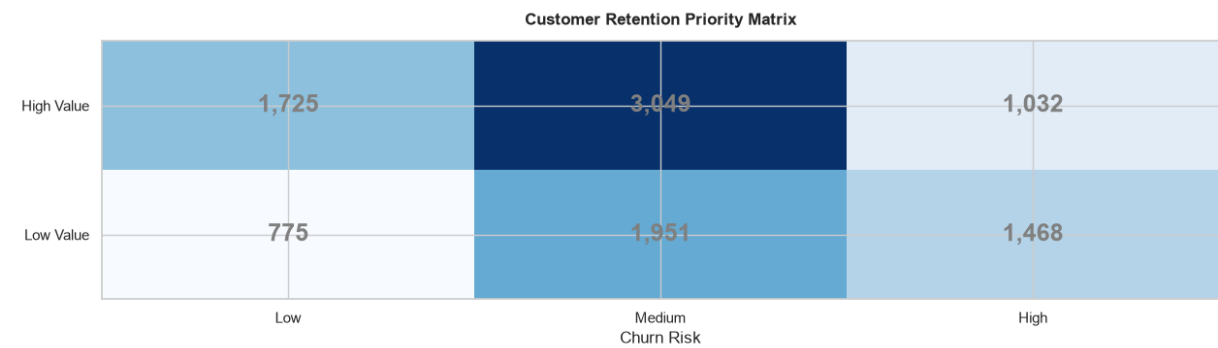


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